nanoker

CERAMIC TECHNICAL DATA SHEET

KZCN

Zirconia-based Composite

KEY PROPERTIES

- High chemical stability
- High corrosion resistance
- Good thermal shock resistance
- Good wear resistance
- Stable at high temperature

APPLICATIONS

- Furnace parts
- Melting crucibles
- Friction wear parts

MAIN PROPERTIES

Property	Value	Unit
Density	3.5	g/cm^3
Fracture Toughness	3.0	$MPa\cdot\sqrt{m}$
Young Modulus	205	GPa
Hardness	11	GPa

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1500	ºC
Recomended temp	<1400	ºC
Continuous temp	1300	ºC

ELECTRICAL BEHAVIOR

Insulator

MATERIAL INFORMATION

Material type	Zirconia-based composite
Structure	Multi-phase ceramic
Key feature	Chemical stability
Machinability	Medium
Composition	Al_2O_3 (50%) / ZrO_2 (30%) / SiO_2 (<20%)

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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